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GODSHAND Kit — Build Map & Source of Truth

> 🟠 **Correction to my earlier note on this page — you were right.** My first pass said "every file is byte-identical, nothing was lost." That was wrong, and I am fixing the record. The core rendering **engine** (all of `src/`, every workflow, `run.py`, the configs) *is* byte-for-byte identical in every build — that part is safe. **But the builds are NOT all equal:** two builds dropped whole features, and one build's launcher was silently **regressed** to an inferior version. Your instinct that "we lost more than the folder" was correct.

✅ What is safe (identical in every build)

The actual engine never changed. These are byte-for-byte identical across FINAL_, TUNED_AK, and FINAL_AK:

- All of `src/` (`render_motion.py`, `forge_character.py`, `voice_pass.py`, `score_gen.py`, `stitch.py`, `postprocess.sh`, …)
- All `workflows/*.json`, `run.py`, and every `config_*.yaml` / `shots_*.json`
- All character reference images in `refs/<name>/face` and `/body`

Nothing about how your film actually renders was corrupted. The damage is only in the **launchers, docs, and scaffolding** that wrap the engine.

⚠️ Feature audit — what was DROPPED or OVERRIDDEN (25 Jul 2026)

Compared file-by-file by content hash, with packaging nesting normalized.

Feature	FINAL_ (your Downloads)	TUNED_AK	FINAL_AK
:--	:-:	:-:	:-:
`NEW_STORY_TEMPLATE/` (author any new film)	✅	❌ dropped	❌ dropped
`GODSHAND_ONECLICK.sh` (main launcher)	✅ full (10,565 B)	❌ dropped	⚠️ **regressed** (9,120 B)
`README_L4.md`	✅ full (3,437 B)	❌ dropped	⚠️ **regressed** (3,129 B)
`l4_model_setup.py` (WAN 2.2 downloader)	✅	❌ dropped	✅
`notebooks/` (2 Kaggle notebooks)	✅	❌ dropped	✅
`RUN_TUNED.sh` (one-command tuned launcher)	❌	✅ only here	❌

🚨 The important one: the launcher was REGRESSED in FINAL_AK

`GODSHAND_ONECLICK.sh` exists in two versions. **The good one is in your Downloads (FINAL_).** The FINAL_AK copy is 1,445 bytes smaller and was rewritten in a way that reintroduces the exact "wrong folder" behavior you have been fighting:

Good version (FINAL_ / Downloads):

- Story is **selectable** — `KIT_CONFIG` picks any film (not just the confrontation).
- Characters are read **dynamically** from the config's `character_forge.targets`.
- Character photos are **OPTIONAL** — if `kit/refs/<name>/face` has images it uses them, otherwise it **auto-forges** a consistent face, logging the choice per character.
- `GODSHAND_ANCHORS` images are copied into `kit/refs/` (the correct home).
- Explicitly states: *"There is no `kit/anchors/` folder — the real location is `kit/refs/<name>/`."*

Regressed version (FINAL_AK):

- **Hardcoded** to the confrontation film + "26 shots" — loses the any-story engine.
- Drops the per-character auto-forge detection.
- Reintroduces a phantom `kit/anchors/` folder and dumps anchor images into `ComfyUI/input/` and the **kit root** — this is the stray-image / wrong-location behavior that caused the Spider-Man / Joker contamination.
- Its `README_L4.md` tells you to use the non-existent `kit/anchors/`.

➡️ **Use the launcher and README from FINAL_ (Downloads). Do not use the FINAL_AK versions.**

✅ Which build to use (corrected)

`GODSHAND_FINAL_` **in your Downloads is your MASTER build.** It has the correct launcher, `NEW_STORY_TEMPLATE/`, the notebooks, `l4_model_setup.py`, and the full README — everything, in its correct form.

The only thing it lacks is `RUN_TUNED.sh` (one-command tuned launcher), and you do not strictly need it — its tuned values are already baked into `NEW_STORY_TEMPLATE/config_newstory.yaml` (`denoise: 0.55`, `codeformer_fidelity: 0.5`, `active_profile: lightning_l4`). To also get the one-command launcher, copy it in:

```
cp GODSHAND_TUNED_AK/RUN_TUNED.sh GODSHAND_FINAL/kit/
```

> Correction: I earlier told you to use `TUNED_AK` as the "richest" build. That was wrong — TUNED_AK actually dropped the launcher, `l4_model_setup.py`, the notebooks, and `NEW_STORY_TEMPLATE/`. FINAL_ (Downloads) is the right master.

🕷️ Root cause — why Spider-Man / Joker appeared as “jack” (the original question)

Three things had to line up. All three are now confirmed in the code:

1. **Two stray images were sitting in jack’s staging folder.**
   `refs/jack/_inbox/jack_preface.jpg` is a Spider-Man (Tobey Maguire) still; `refs/jack/_inbox/joker.png` is a Joker still. `_inbox/` is a “dump a pile of candidate

photos here" staging area that `cull_photos.py` is meant to sort into `face/` and `body/` — but it was never run on jack, so the strays stayed.

2. **The reference loader reads recursively.** `comfy_client.py` and `render_motion.py` collect refs with `glob("refs/jack/**/*", recursive=True)` — so they reach **into `_inbox/`** instead of reading only `refs/jack/face/`.
3. **`_inbox` sorts first, and only the first 5 refs are used.** An underscore sorts before letters, so the recursive list starts with the two `_inbox` images. The code keeps `refs[:5]`, so **Spider-Man and Joker became jack's top-two identity anchors**, ahead of his real face.

And the safety net failed too: the identity-check gate (`_identity_score`) uses the *same* recursive glob, so it scored rendered frames against Spider-Man as if that were jack — it did not just miss the drift, it **approved** it.

Net effect: the blueprint still was conditioned on Spider-Man/Joker → jack animated as Spider-Man → the QC gate rubber-stamped it.

The fix

- **Immediate (done):** `GODSHAND_MASTER.zip` has every `refs/*/_inbox/` folder deleted, so the recursive glob can only find real face/body images.
- **Permanent (one-line, recommended):** restrict the ref globs to the face/body folders, never the parent. In `comfy_client.py`, `render_motion.py` (`_identity_score`), and `refs_for()`, read `refs/<char>/face/*` (and `/body/*`) instead of `refs/<char>/**/*`. Then even if junk lands in `_inbox` again, it can never reach a render.

📁 Where the key things live (never search blind again)

- **Main launcher (correct version):** `GODSHAND_ONECLICK.sh` — in FINAL_ / your Downloads
- **Tuned launcher:** `RUN_TUNED.sh` — TUNED_AK only
- **Model download:** `l4_model_setup.py`
- **Author a new film:** `NEW_STORY_TEMPLATE/` (two files: a shots JSON + a config YAML)

- **Confrontation story + render config:** `config_confrontation.yaml` + `shots_confrontation.json`
- **Character images (real):** `refs/<name>/face/` and `/body/` — cast: `jack, julie, kunal, kunal_40s, kunal_young, gunja, observer, original_identity`
- **The correct anchor location is `kit/refs/<name>/` — there is NO `kit/anchors/` folder.**
- **Junk to delete:** `refs/<name>/_inbox/` — the stray Spider-Man / Joker / Scarlett placeholders live here. The clean design never reads from here.

🎛️ Tuning baked into `RUN_TUNED.sh`

> kaggle_2xt4 profile (2× Tesla T4 — Kaggle has NO L4) · `KIT_MOTION_DENOISE=0.55` (motion freedom, was 0.25) · `KIT_FLF2V_ENDFRAME=1` (shot-to-shot morph ON) · `KIT_FACE_ENHANCE=0` (skip plasticky face composite) · softened temporal denoise · codeformer 0.9 → 0.5

Renders 4 proof shots: `confr_caught`, `confr_allblades`, `confr_dare`, `confr_dontouch`.

🔒 On the sandbox (why your originals were never at risk)

I only ever held a read-only, throwaway copy of the kits you uploaded. I never had write access to your Kaggle kit, your datasets, or your machine. Any differences above already existed in the builds you sent me — I did not create them.

🗒️ Change Log (newest first)

- **25 Jul 2026, ~7:57 AM IST — Fixed the recurring L4 mistake — the proof render now targets Kaggle 2× Tesla T4, not L4 (which Niloy does not have).** `config_confrontation.yaml` and `NEW_STORY_TEMPLATE/config_newstory.yaml` were both pinned to `active_profile: lightning_l4`; switched both to `kaggle_2xt4` (num_gpus: 2, Q8, xformers, 1280×720, rate ₹0). Added a `KIT_PROFILE` env override in `src/cfgutil.py` so the hardware profile can be forced explicitly (prevents silent wrong-GPU runs). Rewrote `RUN_TUNED.sh` for Kaggle 2×T4 (sets `KIT_PROFILE=kaggle_2xt4`, removed the Lightning-only `sudo shutdown` billing step, replaced with a Kaggle 'Stop session' note). Verified no config points at L4

anymore; `cfgutil.py` passes `py_compile`. Reshipped `GODSHAND_MASTER.zip` (239 files).

- **25 Jul 2026, ~7:52 AM IST — Enforced "everything from the master folder."** Verified in code that every script roots to the master folder (`root = parent of src/`) and reads all inputs from there — story (`shots_file`), characters (`refs/`), workflows, grade targets. The only outside doorways are the *opt-in* `GODSHAND_ANCHORS` launcher copy (copies INTO `kit/refs/`) and `lipsync.py`'s lookup for the external LatentSync tool repo (not identity). Hardened `preflight.py` (already stage 1/10 of every run) to (a) print `MASTER FOLDER: <root>` and "all inputs read ONLY from this folder," and (b) warn about any stray ref image outside `face/ body/ costume/`. Reshipped `GODSHAND_MASTER.zip` (239 files); `py_compile` clean.
- **25 Jul 2026, ~7:48 AM IST** — Applied the **permanent code fix** for the Spider-Man bug and reshipped `GODSHAND_MASTER.zip` (238 files). Patched `comfy_client.py._first_image()` (the real render-conditioning leak — it fed the first image, which was the `_inbox` Spider-Man) and `render_motion.py._identity_score()` (the QC gate) to **skip any underscore-prefixed staging folder** (`_inbox`) while still reading `face/` and `body/`. Both files pass `py_compile`; behaviourally proven — with junk dropped back into `_inbox`, the loader now picks `body/jack_body_0.jpg` instead of `_inbox/spiderman.jpg`. Junk can no longer reach a render even if it reappears.
- **25 Jul 2026, ~7:25 AM IST** — Traced and documented the **Spider-Man/Joker root cause** in code: recursive ref glob (`refs/jack/**/*`) reaching into `_inbox/`, `_inbox` sorting first, `refs[:5]` keeping the strays, and the identity-check gate using the same contaminated glob so it approved the drift. Added the "why Spider-Man appeared" section plus the permanent one-line fix (glob `refs/<char>/face/*`, not the parent).
- **25 Jul 2026, ~7:22 AM IST** — Built the clean master kit `GODSHAND_MASTER.zip` (237 files) and delivered it to you as a download. Base = FINAL_ (correct full launcher, 10,565 B); added `RUN_TUNED.sh`; removed every `refs/*/_inbox/` folder (the stray Spider-Man / Joker / Scarlett images). Verified cast faces intact (jack 5, julie 5, kunal 7, observer 5) and `NEW_STORY_TEMPLATE/`, `l4_model_setup.py`, and the notebooks all present.
- **25 Jul 2026, ~7:15 AM IST** — Ran a full file-by-file audit (content hashes, nesting normalized). **Corrected** the earlier "nothing was lost" claim on this

page. Documented: `NEW_STORY_TEMPLATE/` dropped from TUNED_AK & FINAL_AK; launcher dropped from TUNED_AK and **regressed** in FINAL_AK; `l4_model_setup.py` and `notebooks/` dropped from TUNED_AK; `README_L4.md` dropped from TUNED_AK and regressed in FINAL_AK; `RUN_TUNED.sh` only in TUNED_AK. Changed the recommended master build from TUNED_AK → **FINAL_ (Downloads)**.

- **25 Jul 2026, ~7:00 AM IST** — Created this page as the durable build map after the four kit builds were first compared.

🔬 GODSHAND Engine Audit — Code + Real Run Evidence (25 Jul 2026)

📋 Session Log — 25 Jul 2026, Evening (End-to-End Audit Completion)